

DALBANDIN, Pakistan

WMO: 417120

Lat: 28.883N

Lon: 64.400E

Elev: 850

StdP: 91.52

Time Zone: 5.00 (E05)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.9	1.0	-12.8	1.4	13.7	-10.9	1.6	10.4	6.7	19.2	5.1	15.9	0.7	50	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	44.9	21.8	43.9	21.4	42.9	21.0	25.4	40.6	23.9	39.6	22.8	39.3	2.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	17.1	36.7	18.6	14.9	36.4	16.6	13.1	35.9	83.1	40.5	76.4	39.7	71.8	39.4	34.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.3	4.1	3.3	-3.8	47.0	2.7	1.0	-5.8	47.7	-7.4	48.3	-8.9	48.9	-10.9	49.6		
(5)			-6.2	26.9	1.8	3.3	-7.5	29.2	-8.5	31.2	-9.5	33.0	-10.8	35.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.4	11.3	14.3	20.1	26.3	31.7	35.3	36.3	33.9	30.0	24.3	17.0	11.6		
	DBStd	9.48	3.76	4.90	4.04	3.41	2.72	1.85	1.67	1.83	2.50	3.28	3.90	3.55		
	HDD10.0	74	31	18	0	0	0	0	0	0	0	0	2	24		
	HDD18.3	657	218	127	31	1	0	0	0	0	0	1	70	210		
	CDD10.0	5322	71	137	312	488	672	759	814	741	601	443	212	73		
	CDD18.3	2864	1	13	84	239	413	509	556	483	351	185	31	1		
	CDH23.3	46197	27	180	1047	3160	6519	8943	9929	8151	5276	2456	473	36		
	CDH26.7	31201	1	40	414	1783	4333	6561	7456	5761	3416	1284	148	4		
(14) Wind	WSAvg	1.3	1.2	1.3	1.4	1.3	1.4	1.5	1.4	1.3	1.1	1.1	1.1	1.1		
(15) Precipitation	PrecAvg	83	15	15	18	7	2	1	4	1	0	2	3	9		
	PrecMax	215	72	61	85	47	14	9	87	18	3	24	43	82		
	PrecMin	10	0	0	0	0	0	0	0	0	0	0	0	0		
	PrecStd	49	18	18	21	11	3	2	13	3	1	4	6	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.6	30.2	35.1	39.9	44.2	46.2	46.4	44.9	41.6	39.1	32.4	26.9		
		MCWB	12.8	15.1	16.8	18.3	21.3	21.4	21.9	23.2	19.5	19.2	15.5	13.1		
	2%	DB	23.8	27.7	32.7	37.7	42.6	44.9	45.0	43.3	40.5	36.9	30.3	24.1		
		MCWB	12.4	14.1	16.4	18.4	20.6	21.5	21.8	21.3	19.2	17.3	15.7	12.7		
	5%	DB	21.8	25.6	30.8	36.2	41.1	43.7	44.1	42.2	39.5	35.0	28.4	21.9		
		MCWB	11.3	13.3	15.4	18.3	19.9	21.1	21.5	20.7	18.8	16.8	14.7	11.3		
	10%	DB	19.7	23.4	29.0	34.6	39.6	42.5	43.1	41.1	38.3	32.9	26.1	19.7		
		MCWB	10.1	12.4	14.7	17.5	19.2	20.7	21.5	20.0	18.3	16.1	13.6	10.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.3	16.0	19.0	21.7	25.7	26.1	28.0	26.9	22.8	21.0	18.6	14.7		
		MCDB	23.3	28.1	31.8	36.3	41.8	42.3	42.6	41.1	37.3	35.5	30.1	23.2		
	2%	WB	13.1	14.7	17.4	20.3	22.4	24.0	25.9	24.3	21.4	19.4	16.6	13.1		
		MCDB	21.9	25.9	29.7	34.0	38.2	41.1	40.0	39.5	37.6	33.0	27.6	22.7		
	5%	WB	12.1	13.9	16.4	19.1	20.9	22.6	24.1	23.1	20.2	18.1	15.4	11.9		
		MCDB	20.6	24.5	28.5	33.8	38.5	40.8	39.5	37.6	37.7	32.5	26.3	20.2		
	10%	WB	10.9	12.8	15.5	18.1	19.7	21.5	23.0	21.9	18.9	16.9	14.2	10.8		
		MCDB	18.4	22.3	27.1	32.8	37.6	40.6	40.1	37.5	36.9	31.7	25.2	18.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.8	14.3	14.7	14.7	14.3	14.5	13.9	14.6	16.5	16.8	16.5	15.7		
		MCDBR	17.7	16.5	16.7	16.4	16.0	15.8	14.9	15.7	17.2	17.7	18.5	18.2		
		MCWBR	9.6	7.8	7.0	6.7	6.5	6.0	5.8	6.1	7.1	7.5	9.5	9.7		
	5% WB	MCDBR	15.3	15.6	15.7	15.5	15.2	15.2	13.8	13.5	16.6	16.9	17.2	16.5		
		MCWBR	8.7	7.9	7.7	7.3	7.1	6.8	6.6	6.3	7.9	8.9	9.7	9.2		
(40) Clear-Sky Solar Irradiance	taub		0.351	0.410	0.457	0.517	0.532	0.530	0.604	0.541	0.434	0.398	0.359	0.334		
	taud		2.272	2.086	1.941	1.830	1.797	1.803	1.680	1.790	2.001	2.095	2.241	2.321		
	Ebn at Noon		876	847	829	793	780	777	720	763	839	846	858	879		
	Edh at Noon		114	149	184	212	220	218	245	218	170	145	116	103		
(44) All-Sky Solar Radiation	RadAvg		3.92	4.62	5.83	6.77	7.43	7.79	7.26	7.08	6.61	5.57	4.43	3.79		
	RadStd		0.24	0.32	0.25	0.31	0.22	0.32	0.35	0.27	0.17	0.12	0.14	0.18		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page